Efficacy of Home Glucose Monitoring in Diabetic Pregnancy

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A randomized, prospective study was performed to evaluate the efficacy of daily home glucose monitoring on the outcome of pregnancies in women with insulin-dependent diabetes mellitus. Home glucose monitoring was compared with a weekly venipuncture protocol. No differences were observed between groups in clinical features (age, parity, White's classification) or representative delivery outcomes (method of delivery, weeks' gestation, or weight at birth). No statistically significant differences were observed between the groups in several aspects of perinatal morbidity. However, home glucose monitoring was associated with fewer readmissions for diabetic control ($p = 0.05$), fewer days in the hospital ($p < 0.01$), and decreased total patient expense ($p < 0.05$).

In the preceding decade, there have been remarkable improvements in the management of pregnancy complicated by insulin-dependent diabetes mellitus. The widespread use of fetal pulmonary maturity studies and modern tests of fetal well-being have helped to decrease perinatal morbidity and mortality rates. In the past few years, considerable effort has also been directed toward normalization of maternal glucose levels. This has been effected both by altering the frequency and route of insulin administration and by increasing the frequency of blood glucose measurements. Several new instruments have been developed that allow patients to determine their own blood glucose levels at home. Although several reports have documented the safety of these devices during pregnancy [1–3], their relative efficacy compared with that of traditional venipuncture protocols has not been documented. The purpose of this paper is to report the maternal and neonatal outcomes of a prospective randomized trial of patient glucose monitoring versus a conventional outpatient management protocol.

PATIENTS AND METHODS

Patients. Between February 1, 1980, and September 16, 1981, pregnant insulin-dependent diabetic women were recruited for participation from the High-Risk Obstetric Clinic at the University of Iowa Hospitals and Clinics. All women required insulin before conception, and all were at less than 20 weeks' gestation when initially enrolled. A total of 34 women were recruited, and 30 agreed to participate, representing a successful recruitment rate of 88 percent.

Protocol. Consenting patients were assigned to control and experimental groups using a random number sequence. Patients in the control group were managed in accordance with the extant standard of care for pregnant diabetic patients at our institution. All patients were admitted after the first clinic visit for metabolic control (aimed at attaining a fasting serum glucose level of 70 to 110 mg/dl and a two-hour postprandial serum glucose level of 80 to 130 mg/dl) and baseline evaluation (24-hour urine collection for total protein level

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and creatinine clearance levels, SMA-12 analysis, ophthalmologic evaluation, electrocardiography, and hemoglobin A_{1C} determination). The average duration of amenorrhea in the control patients at the time of initial admission was 9.3 ± 1.4 weeks. Patients began receiving a three-meal/three-snack American Diabetes Association diet based on 35 kcal/kg ideal body weight. Subcutaneous insulin was administered twice daily as combinations of regular plus intermediate-acting (lente or NPH) forms. When metabolic control was achieved, the patients were discharged to be followed in the High-Risk Obstetric Clinic. Patients were routinely seen every two weeks until 32 weeks' gestation and weekly thereafter. Serum glucose levels (fasting, two hours after breakfast, and two hours after lunch) were measured on one day each week, and the insulin dosage was adjusted accordingly. Patients were instructed to telephone on a weekly basis or as needed to report blood sugar levels or possible complications. Antepartum care of all patients was supervised by one physician. Although no strict criteria for readmission were enforced, patients were generally readmitted if two or more serum glucose values exceeded the limits just outlined. An ultrasound examination of the fetus and maternal serum alphafetoprotein determination were performed at 16 to 18 weeks' gestation. Maternal hemoglobin A_{1C} levels were determined at the initial visit and every six to eight weeks thereafter. Fetal surveillance was begun at 28 to 32 weeks on at least a weekly basis until the predelivery admission and then more extensively during hospitalization. All patients were admitted for the duration of the pregnancy by the completion of 36 weeks' gestation.

Patients in the experimental group were also admitted at the first clinic visit for metabolic control and baseline evaluation as previously outlined. The average duration of amenorrhea in the experimental group at the time of initial admission was 10.0 ± 2.1 weeks. During this admission, patients were instructed in the operation of a home monitoring system employing glucose oxidase-impregnated reagent strips for whole blood glucose determination (Dextrometer, Ames Company, Elkhart, Indiana). Within 72 hours, all patients demonstrated satisfactory proficiency in the system's use, by comparison with simultaneous specimens submitted to the clinical laboratory. When adequate metabolic control was achieved, these patients were discharged and also followed in the High-Risk Obstetric Clinic. All patients' visits were supervised by the same physician. These patients were also instructed to telephone at least weekly to report their blood glucose values. Fasting plus two-hour postprandial morning, afternoon, and evening glucose values were monitored daily by the patients. Insulin dosage was adjusted by the patient only after physician consultation. Readmission procedures, ultrasound study, alphafetoprotein and hemoglobin A_{1C} determinations, fetal surveillance, and predelivery admission criteria were identical to those employed in the control population.

One patient randomly assigned to each group had a first-trimester spontaneous miscarriage. Both of these patients had hemoglobin A_{1C} values (7.8 percent and 7.6 percent) near the population average. Their hypoglycemic episodes during their initial hospitalization (four and five) was also similar to those in the entire population (control population mean 3.8 ± 2.2, experimental population mean 4.1 ± 3.0). Thus, only 14 patients in each original group are further described. Method of delivery, weeks' gestation (as determined by menstrual history, ultrasound study, and Ballard gestational age assessment examination [4]), and weight at birth were analyzed for each of the remaining 28 patients.

TABLE I Clinical Data

	Experimental Group	Control Group
Number	15	15
Average age (years)	24.0 ± 4.0	23.3 ± 4.4
Average parity	1.4 ± 0.7	1.3 ± 0.8
White's class [7]		
B	4	3
C	2	5
D	5	3
F-R	4	4

Hemoglobin A_{1C} determinations were performed using the technique of Goldstein et al [5]. Perinatal morbidity was assessed by polycythemia, hypocalcemia, hyperbilirubinemia, and hypoglycemia. Cord vein serum was assayed for C-peptide using the technique of Heding [6].

The potential economic impact of home glucose monitoring in diabetic pregnancy was evaluated by readmissions for control of diabetes, total days in the hospital, and total patient expense. The latter figure includes all hospital, laboratory, and physician fees for both diabetic patients and their infants. It does not include expenses incurred by patients traveling to and from appointments. Because the Dextrometers are the property of the Department of Obstetrics and Gynecology and are loaned to patients during their pregnancies, their cost is not included. If such costs were to be considered, the average retail value of each instrument is $275 and the cost of supplies per patient per day is approximately $1.50 (four Dextrostix plus calibrating reagents). Precise cost accounting for all telephone calls is not possible, as all calls from physician to patient were made through a Wide Area Telecommunications System. However, the total number of telephone conversations between each patient and physician was recorded. Because all but one patient resided outside the local telephone dialing area, an arbitrary cost of $1.00 per telephone call per patient was assigned to each total patient expense.

Statistical analyses were performed using either the small-sample t test or chi-square analysis. Statistical significance was defined as a p value equal to or less than 0.05.

RESULTS

Clinical data on the two patient subgroups are shown in **Table I.** No statistically significant differences existed between the experimental and control groups in average age, parity, or White's classification [7]. **Table II** documents delivery outcomes for both groups, including method of delivery, average weeks of gestation, and average weight at birth. For the experimental group, the gestational ages ranged from 30 to 40 weeks, and weight at birth was 1,000 to 3,930 g. For the control

TABLE II Representative Delivery Outcomes

	Experimental Group	Control Group
Method of delivery		
Cesarean section	7	9
Vaginal	7	5
Weeks of gestation ± SD	38.0 ± 3.1	37.6 ± 2.4
Range	30–40	33–39
Weight at birth (g) ± SD	2,986 ± 488	3,176 ± 467
Range	1,000–3,930	1,600–3,990

group, the gestational ages ranged from 33 to 39 weeks, and weight at birth was 1,600 to 3,990 g.

There was one perinatal death in this series, a 1,320 g male born at 32 weeks' gestation to a 26-year-old primigravida with class F-R diabetes. The mother had been enrolled in the experimental group at 11 weeks' gestation. The infant was delivered by primary cesarean section because of postive results of an oxytocin challenge test. The infant was noted to have radial aplasia, persistent thrombocytopenia, and severe respiratory distress syndrome. Although the initial respiratory distress syndrome was successfully managed, severe pulmonary hemorrhage developed on the ninth day of life, and the infant died several hours later. This child was also the only neonate with any detectable anomalies in the entire population.

Table II also demonstrates that 16 of 28 patients (57.1 percent) had delivery by cesarean section. In the experimental population, two cesarean sections were performed for fetal distress, two because of a history of previous cesarean section, and three because of failure to progress in labor. In the control population, one cesarean section was performed for fetal distress, three because of a history of previous cesarean section, four because of failure to progress in labor, and one because of breech presentation in a primigravida at 40 weeks' gestation.

As an attempt to assess the degree of diabetic control during pregnancy, hemoglobin A_{1C} assays were performed at the initial visit and then at six- to eight-week intervals. The results are shown in **Figure 1.** No statistically significant differences existed between the home monitoring and control groups at any stage of pregnancy.

The effects of the two antepartum management protocols on indexes of perinatal morbidity are documented in **Table III.** No statistically significant differences existed between the groups for any of the four parameters. Cord vein C-peptide values did not differ between the groups (control 0.93 ± 0.90 ng/nl, experimental 1.06 ± 0.80 ng/nl).

Relative economic benefits were assessed using the parameters listed in **Table IV.** As mentioned, total patient expense in the home monitoring group does not include the cost of the home monitor or its supplies. The documented telephone consultations in the two groups of patients were not significantly different (control mean 11.2 ± 5.2, range 3 to 22; experimental mean 13.0 ± 3.4, range 9 to 19; $p > 0.05$).

COMMENTS

The study population was recruited from consecutive insulin-dependent diabetic patients during the 19.5 month recruitment period. Because 30 of 34 recruited patients consented to participate, the clinical data in Table I are representative of the pregnant diabetic population in our institution during recent years. The preponderance of patients in White's classes D and F-R (16 of 30) reflects the referral basis of our High-Risk Obstetric Service.

As noted in Table II, in 16 of 28 viable pregnancies (57.1 percent), delivery was abdominal. This rate is

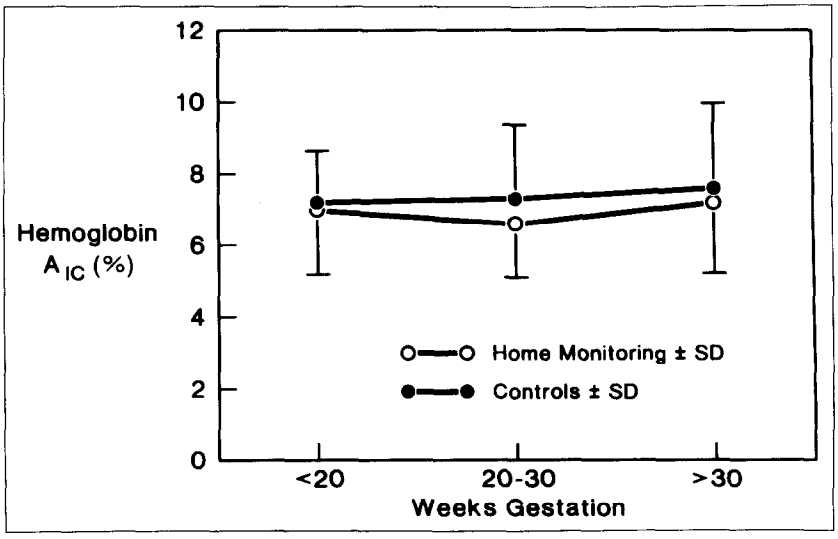

Figure 1. Hemoglobin A_{1C} average values ± standard deviation in home monitoring (O—O) and control (●—●) groups. The results at less than 20 weeks' gestation represent only initial values at the time of recruitment into the study. Values at 20 to 30 and more than 30 weeks represent mean ± standard deviation levels for all assays during those time periods.

TABLE III Perinatal Morbidity

	Experimental Group	Control Group
Polycythemia*	0/14	1/14
Hypocalcemia*	1/14	1/14
Hyperbilirubinemia*	5/14	9/14
Hypoglycemia†	4/14	7/14

* Defined as requiring therapy.
† Defined as a true serum glucose of less than 30 mg/dl.

substantially higher than the overall cesarean section rate at our institution (13.0 percent in calendar 1981). Of note, six of nine patients in White's F-R class had abdominal delivery, including all three cesarean sections performed for fetal distress. This suggests that fetuses of women with diabetic microangiopathy do not tolerate labor well. Although it is tempting to ascribe the low average birth weight to good control of diabetes, the preponderance of patients in White's classes D and F-R may in part also explain this finding. It should be noted that the average duration of pregnancy in both populations extends to within two to under two and a half weeks of the expected date of confinement. This reflects progressive confidence in fetal surveillance techniques, plus appreciation of the potential morbidity associated with delivery more than two weeks before the expected date of confinement.

No statistically signficant differences were noted in perinatal morbidity indexes between the two study groups (Table III). However, there is a suggestion that hyperbilirubinemia and hypoglycemia may be less frequent in the home monitoring group. A larger study population might document signficant differences.

Several authors have documented the safety of home glucose monitoring devices during pregnancy [1–3]. The current study confirms those observations. In our two similar study populations (Table I), no sigificant differences were noted in hemoglobin A_{1C} values through pregnancy (Figure 1), representative delivery outcomes (Table II), or neonatal morbidity (Table III). However, home glucose monitoring during diabetic pregnancy did have a definite economic impact, as noted in Table IV. Patients monitoring their glucose level daily at home were less likely to require readmission for control of diabetes. During the course of this study, it became obvious that home glucose monitoring also had a substantial educational benefit to patients and their families. By identifying those patterns in their daily home environment that disrupted diabetic control, patients could not only avoid prolonged hospitalizations, but also learn the importance of careful attention to diet, activity, and insulin administration. These educational benefits, when combined with patients' feelings of more active participation in their care and management, resulted in the substantial subjective impression that

TABLE IV Economic Parameters

	Experimental Group	Control Group
Readmission for control of diabetes	8 ($p = 0.05$)	27
Range	0–2	0–5
Total hospitalization mean ± SD (days)	28.2 ± 17.6 ($p < 0.01$)	42.0 ± 20.0
Range	14–67	17–89
Total pregnancy expense mean ± SD (dollars)	9,471 ± 4,804 ($p < 0.05$)	14,662 ± 6,412
Range	4,677–16,238	6,462–29,851

patients in the home monitoring group were more enthusiastic about both the pregnancy and subsequent postpregnancy diabetic management. This has been reflected by the fact that 12 of the 14 patients in the home monitoring group have continued a home blood glucose monitoring program at their own expense after the pregnancy. This study was not doubleblinded, and it could, therefore, be argued that observer bias distorted the relative readmission frequency. On the other hand, the data could be interpreted as demonstrating that moderate blood sugar abnormalities can be managed successfully on an outpatient basis, given the capability of following blood glucose values on a daily basis.

Table IV demonstrates the average savings in total days of hospitalization and patient expense. Between the two study groups, it represents a difference of 193 days of hospitalization and a savings in hospital, laboratory, and physician-service expenses of $72,674. Similar economic benefits have also been reported in a recent retrospective study [3].

Although not addressed in this study, the frequency of clinic visits may possibly be decreased using such a patient management system. Especially in a nonurban setting such as that of the University of Iowa, this could result in additional patient savings, since many patients must travel several hundred miles to and from each clinic visit.

In summary, in this small, prospective study, home glucose monitoring was not associated with any obvious increased maternal or perinatal risks. Its use has generated significant enthusiam among this patient population. It has also resulted in substantial economic advantage and it is now being employed on a routine basis on our service.

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